

Health Hub

Application Documentation

1. User Roles

The platform supports six portal types. Role tags are used internally to route users to the correct portal; sub-roles within the same area share a single interface.

Role	Portal Access
Patient	Patient Portal
Health Expert (GP)	Health Expert Portal
Specialist	Specialist Portal
Pharmacist	Pharmacy Portal
Diagnostics	Diagnostics Portal
Admin	Admin Portal

2. Patient Portal (/patient/*)

- Dashboard** — Home view with key metrics and quick actions
- Appointments** — View all scheduled consultations
- Video Consultation** — Join a live video call with a Health Expert or Specialist
- Medical Records** — Personal health history
- Waiting Room** — Queue status while waiting for a consultation
- Notifications** — Alerts and updates from the care team
- AI Health Chat** — Symptom checker with GP booking and HealWell content recommendations
- Profile** — Personal details and billing tab — add or remove payment cards
- HealWell Africa / HealWell India** — Regional health and wellness content
- Travel Health** — Travel vaccinations and related services
- Patient Pharmacy** — View and track prescriptions
- Patient Specialist** — View and manage specialist referrals

3. Health Expert Portal (/gp)

- Dashboard / Queue** — Live consultation queue with accept and reject controls
- Consultation History** — Record of past consultations
- Manage Consultation** — Mark consultations as complete or remove them
- Profile** — Credentials, specialty, and online/offline status toggle

4. Specialist Portal (/specialist/*)

Dashboard — Incoming referrals list

Referral Details — Accept, schedule, request additional information, or reassign a referral

Specialist Consultation — Run the consultation — chat, prescribe, and order lab tests

Profile — Specialty, facility, and credentials

5. Pharmacy Portal (/pharmacy/*)

Prescription Scanner — Scan or look up a prescription by code

Prescription Details — View, claim, and dispense a prescription

Dispensing History — Full log of dispensed prescriptions

Profile — Pharmacy credentials

6. Diagnostics Portal (/diagnostics/*)

Lab Orders — Incoming orders organised by status tabs

Order Details — Full detail view for a single order

Result Upload — Upload test results as PDF or image

Profile — Facility and lab credentials

7. Admin Portal (/admin/*)

Dashboard — System overview and key metrics

User Management — List users, create accounts, update roles and status, approve provider access requests

Activity Logs — Full record of user activity across the platform

Service Requests — Handle travel and callback requests submitted by patients

Prescriptions — View all prescriptions and contact patients where required

Diagnostics — View all lab orders and contact patients where required

Referrals — View all referrals and manage workflow progression

System Health — Server health metrics and uptime monitoring

8. Authentication (/auth/*)

Login — Secure login with rate limiting

Signup — Registration for patients and providers

Forgot Password — Password reset flow

Provider accounts (Health Expert, Specialist, Pharmacist, Diagnostics) require manual approval by an Admin before access is granted.

9. Core Workflows

9.1 Patient Consultation

1. Patient submits a consultation request
2. Health Expert sees the request in the queue and accepts
3. A video room is created and both parties are notified
4. Consultation takes place via video call
5. Health Expert marks the consultation as complete
6. Patient receives a summary notification

9.2 Referral

1. Health Expert creates a referral to a Specialist
2. Specialist receives a notification and reviews the referral
3. Specialist accepts and schedules a consultation, or requests additional patient information
4. Consultation takes place via video call
5. Referral is marked as complete

9.3 Prescription and Pharmacy

1. Health Expert or Specialist creates a prescription
2. Pharmacy scans or looks up the prescription code
3. Pharmacy claims the prescription
4. Pharmacy dispenses the prescription
5. Patient receives a pickup notification
6. Admin receives a copy and contacts the patient if any follow-up is needed

9.4 Diagnostics

1. Health Expert or Specialist raises a lab order
2. Diagnostics team receives the order and marks it as received
3. Results are uploaded to the platform
4. Health Expert and patient are notified that results are ready
5. Admin receives a copy and contacts the patient if any follow-up is needed

9.5 AI Health Triage

1. Patient chats with the Healthcare AI and describes symptoms
2. AI works through triage questions to assess severity
3. Based on severity: patient books a Health Expert, watches a wellness video online, or uses HealWell at home

9.6 Payments

1. Patient adds a payment card via the billing tab in their profile
2. Card is saved securely for future use
3. A charge is raised when a consultation or service is confirmed
4. Payment confirmation is sent to the patient

10. Integrations

Integration	Purpose
Daily.co	Video consultations between patients and providers
Stripe	Payment processing and card management
Healthcare AI	Symptom triage and care pathway recommendations
Redis	Real-time messaging via WebSocket
PostgreSQL	Primary data store

11. Real-Time Communication

The platform uses WebSocket connections to push live updates to connected users. Channels are scoped by role and by individual user, so each party receives only the events relevant to them.

- Consultation updates** — Status changes and new messages during a consultation
- Referral status changes** — Notifications when a referral is accepted, scheduled, or completed
- Lab orders** — New orders and result uploads for the Diagnostics team
- Prescription dispensing** — Confirmation when a prescription has been dispensed
- Service request updates** — Admin responses to travel and callback requests

The WebSocket layer is backed by Redis pub/sub, which allows the platform to scale across multiple server instances without losing message delivery.